SCHOOL OF ENGINEERING, CUSAT

B. TECH DEGREE 2nd SEMESTER SECOND INTERNAL EXAMINATIONS, JULY 2024

23-200-0205B INTRODUCTION TO CYBER PHYSICAL SYSTEMS

(CSA, CSB, EEA, EEB, ITA, ITB, ECA, ECB)

(2023 Scheme)

TIME: 2 hrs **MAX. MARKS: 30**

Course Outcomes

On completion of this course the student will be able to:

CO1	Understand the features & components of Cyber Physical Systems
CO2	Understand the elementary constructs of Arduino Software
CO3	Develop optimal programs & circuits for interfacing various sensors with Arduino
CO4	Apply Arduino IDE for developing suitable programs for interfacing sensors & actuators with Arduino & Node MCU

PART A **(5 x 2= 10 Marks)**

(Answer all questions)

Q.No.	Questions	Marks	BL	CO	PO
I (a)	What functions are used to setup serial communication in Arduino?	2	L3	2	1
(b)	How does the analogRead() function work in Arduino?	2	L2	2	1
(c)	How can you configure a pin as an input or output in Arduino?	2	L3	2	3
(d)	What function is used to connect NodeMCU to a WiFi network?	2	L3	4	2
(e)	What is a pull-up resistor, and how is it used in a push button circuit?	2	L2	3	1

PART B
(Answer the following questions choosing either the first OR second option)

Q.No.	Questions	Marks	BL	CO	PO
			(1 x 10 = 10 Marks)		
II (a)	What is PWM, and how is it implemented in Arduino?	6	L2	2	5
(b)	Explain the difference between sinking and sourcing methods in LED interfacing.	4	L4	2	3
	OR				
III (a)	How can you display a value on an LED bar graph using Arduino?	5	L6	3	1
(b)	Explain how to interface an analog temperature sensor with Arduino.	5	L3	1	1
			(1 x 10 = 10 Marks)		
IV (a)	List some key features of the NodeMCU/ESP32	4	L1	4	5
(b)	Explain the interfacing between any motor (DC/Stepper/Servo) with Arduino.	6	L3	4	3
	OR				
V (a)	Write a simple code to blink an LED connected to pin D1 on NodeMCU.	5	L3	4	1
(b)	How do you connect NodeMCU to a WiFi network?	5	L3	4	6

Bloom's Taxonomy Levels

Bloom's taxonomy levels (L1-Remember, L2-Understand, L3-Apply, L4-Analyze L5-Evaluate L6-Create), PO-Program outcome

L1-%	L2- %	L3-%	L4-%	L5-%	L6-%
7	23	53	7	0	7